System Validation Report

End-to-end validation of the running system against the SRS v1.1 requirements – Software Test Plan v1.1, test levels 3–5

Project	WealthWise – Wealth Management System (Team-02)
Report	System Validation Report
Date	02-11-2026
Build under test	main @ a2c6973 (Sprint 2 + final fixes, PR #77) + testing-week tests
Environment	Linux test machine · Node.js 22.22 · PostgreSQL 16.15 · Next.js 14.2.35 · TypeScript 5.9 · c8 10.1.3 (V8 coverage) · Chromium via Playwright
Run	2026-11-02 14:17 UTC, production build (`npm run build && npm start`) on a freshly migrated and seeded database
Method	Scripted HTTP run of every STP test case (actual result recorded and compared with the expected result), extra instances for TC-052/TC-053, Playwright UI checks (TC-051, TC-057), automated database suite for TC-038

STP test cases
59/59
passed

API operations exercised
53/53
every route and method

Requirements validated
42/42
26 FR · 7 NFR · 9 SR

Open defects
0
3 found and fixed

1. Results by area

Area	Test cases	Executed	Passed	Failed
Authentication & access	TC-001 – TC-009	9	9	0
Assets	TC-010 – TC-014	5	5	0
Loans & EMI	TC-015 – TC-025	11	11	0
Ledger & search	TC-026 – TC-028	3	3	0
Import & rules	TC-029 – TC-033	5	5	0
Receipts	TC-034 – TC-035	2	2	0
Vault	TC-036 – TC-038	3	3	0
Dashboard	TC-039	1	1	0
Insights	TC-040 – TC-042	3	3	0
Reports	TC-043 – TC-044	2	2	0
Notifications & news	TC-045 – TC-046	2	2	0
Advisor	TC-047	1	1	0
Security	TC-048 – TC-053	6	6	0
Non-functional	TC-054 – TC-059	6	6	0
Total		**59**	**59**	**0**

By priority

Priority	Executed	Passed	Failed
High	33	33	0
Medium	23	23	0
Low	3	3	0

2. Requirement validation

Every SRS requirement is validated by at least one passed system test case (primary links from the STP).

Requirement	Validated by	Result
FR-01	TC-001, TC-002, TC-003, TC-004	**Pass**
FR-02	TC-005, TC-006	**Pass**
FR-03	TC-007	**Pass**
FR-04	TC-008, TC-009	**Pass**
FR-05	TC-010, TC-011, TC-012, TC-014	**Pass**
FR-06	TC-013	**Pass**
FR-07	TC-015, TC-016	**Pass**
FR-08	TC-017, TC-018	**Pass**
FR-09	TC-019, TC-020, TC-021	**Pass**
FR-10	TC-022	**Pass**
FR-11	TC-023, TC-024, TC-025	**Pass**
FR-12	TC-026, TC-027	**Pass**
FR-13	TC-028	**Pass**
FR-14	TC-029, TC-030, TC-031, TC-032	**Pass**
FR-15	TC-029, TC-033	**Pass**
FR-16	TC-034, TC-035	**Pass**
FR-17	TC-036, TC-037, TC-038	**Pass**
FR-18	TC-010, TC-039	**Pass**
FR-19	TC-040	**Pass**
FR-20	TC-041	**Pass**
FR-21	TC-042	**Pass**
FR-22	TC-043	**Pass**
FR-23	TC-044	**Pass**
FR-24	TC-045	**Pass**
FR-25	TC-046	**Pass**
FR-26	TC-047	**Pass**
NFR-01	TC-054	**Pass**
NFR-02	TC-015, TC-055	**Pass**
NFR-03	TC-025, TC-032	**Pass**
NFR-04	TC-038, TC-056	**Pass**
NFR-05	TC-057	**Pass**
NFR-06	TC-058	**Pass**
NFR-07	TC-053, TC-059	**Pass**
SR-01	TC-048	**Pass**
SR-02	TC-005, TC-049	**Pass**
SR-03	TC-006	**Pass**
SR-04	TC-008, TC-009	**Pass**
SR-05	TC-047, TC-050	**Pass**
SR-06	TC-011, TC-016, TC-027, TC-051	**Pass**
SR-07	TC-031, TC-035, TC-037	**Pass**
SR-08	TC-052	**Pass**
SR-09	TC-044, TC-053	**Pass**

3. API operation coverage

53 of 53 API operations were called during the run (status codes observed in brackets).

Method	Endpoint	Status codes seen
GET	/api/advisor/clients/{id}	200, 403
GET	/api/advisor/clients	200, 403
GET	/api/advisor/me	200, 403
PUT	/api/advisor/me	200, 404
DELETE	/api/advisor/me	200
GET	/api/assets/{id}	200, 404
PUT	/api/assets/{id}	200, 404
DELETE	/api/assets/{id}	200, 404
GET	/api/assets	200, 401
POST	/api/assets	201, 422
POST	/api/auth/login	200, 401
POST	/api/auth/logout	200
GET	/api/auth/me	200
POST	/api/auth/signup	201, 409, 422
GET	/api/dashboard	200, 401
GET	/api/dues	200
POST	/api/emi/{id}/pay	200, 404
GET	/api/health	200
GET	/api/imports/batches	200
POST	/api/imports/commit	201, 422
POST	/api/imports/preview	200, 413, 415, 422
GET	/api/insights/projection	200, 404, 422
POST	/api/insights/recommend	200
GET	/api/insights/score	200
GET	/api/loans/{id}/prepay	200
POST	/api/loans/{id}/prepay	201, 404, 422
POST	/api/loans/{id}/prepay/simulate	200
GET	/api/loans/{id}	200, 404
PUT	/api/loans/{id}	409
PATCH	/api/loans/{id}	200, 422
DELETE	/api/loans/{id}	200
GET	/api/loans/{id}/schedule	200
GET	/api/loans	200, 401
POST	/api/loans	201, 422
GET	/api/news/{id}/impact	200
POST	/api/news/refresh	403, 503
GET	/api/news	200
GET	/api/notifications	200
POST	/api/notifications	200
POST	/api/receipts/confirm	201, 409
POST	/api/receipts/scan	200, 413, 415
GET	/api/reports/fy	200
GET	/api/reports/statement	200, 401
DELETE	/api/rules/{id}	200, 404
GET	/api/rules	200
POST	/api/rules	201

Method	Endpoint	Status codes seen
DELETE	`/api/transactions/{id}`	200, 404
GET	`/api/transactions`	200, 401
POST	`/api/transactions`	201, 422
GET	`/api/vault/{id}/file`	200, 404
DELETE	`/api/vault/{id}`	200, 404
GET	`/api/vault`	200, 401
POST	`/api/vault`	201, 413, 415

4. Supplementary checks

API operations not reached by an STP test case were checked separately.

ID	Check	Actual result	Result
SV-01	Health check reports database connectivity	HTTP 200 {"status":"ok","database":"connected","time":"2026-11-02 14:17:54.638807+00"}	**Pass**
SV-02	Change loan status with PATCH	PATCH defaulted → 200; PATCH 'paused' → 422; DELETE → 200; GET after delete → 404	**Pass**
SV-03	Individual chooses, views and removes an advisor	PUT → 200; GET → advisor@wealthwise.dev; appears in advisor's client list: true; non-advisor email → 404; advisor calling it → 403; DELETE → 200, then advisor null	**Pass**
SV-04	Delete own categorisation rule	POST → 201; DELETE own rule → 200; DELETE again → 404	**Pass**
SV-05	Owner downloads a vault file	HTTP 200; application/pdf; attachment; filename="d.pdf"; nosniff: true; 4096 bytes	**Pass**

5. Non-functional results

Requirement	Target	Measured	Result
NFR-01 performance	Dashboard p90 ≤ 2 s; OCR ≤ 10 s	dashboard p90 9 ms over 20 requests (max 13 ms); OCR scans 0.8 s, 0.7 s, 0.8 s, 0.8 s, 0.8 s	**Pass**
NFR-02 exact money	Σ principal = loan amount; final balance 0.00	(a) EMI for ₹1,20,000 at 0 % over 12 months = ₹10000, no division error; (b) ₹1,00,001 at 7.25 % for 37 months: 37 installments, Σ principal ₹100001, final balance 0	**Pass**
NFR-03 atomicity	No partial change after a mid-operation failure	TC-025, TC-032 and the fault-injection database tests: nothing left behind	**Pass**
NFR-04 degradation	Core works when vault storage / RBI feed fail	with the RBI feed unreachable: asset, loan, EMI payment, transaction, dashboard, import preview and commit → 201/201/200/201/200/200/201; news fell back to cached/sample articles (3 ms, then 3 ms – failed fetch not retried within 60 s); vault storage failure → 503 covered by TC-038	**Pass**
NFR-05 usability	Theme persists, Indian grouping, labelled charts	dark theme kept after reload: true; dark class set before first paint: true; amounts like ₹52,87,014.37: true; every chart has a legend, labels or table: true; ledger shows "<script>" text without running it: true	**Pass**
NFR-06 maintainability	0 type errors, all tests pass, layering	typecheck: 0 errors; unit tests: 73 passed, 0 failed; database tests: 54 passed, 0 failed; layering checked by tests/unit/architecture.test.ts (part of the unit run)	**Pass**
NFR-07 portability	Fresh install from the README steps	Fresh checkout → migrate → seed → build → start; also on every CI run	**Pass**

6. Defects

ID	Severity	Found	Description	Status
DEF-01	High	31-10-2026, TC-033	Route files for `/api/rules` and `/api/rules/{id}` missing on `main`; rules panel could not list, add or delete rules	Fixed in PR #77; retested – Pass (TC-033, SV-04)
DEF-02	Medium	31-10-2026, TC-014	Deleting an asset or loan left transactions pointing at a deleted row	Fixed in PR #77 (migration 011); retested – Pass (TC-014)
DEF-03	Low	31-10-2026, TC-032	Impossible date in an import commit returned 500 instead of 422 (data unaffected)	Fixed in PR #77; retested – Pass (TC-032)

This run (02-11-2026) found no new defects.

7. Observations and limitations

- The live RBI press-release feed was unreachable from the test network; TC-046 validated the fallback path. Feed refresh and parsing are covered by database and unit tests with a stubbed feed.
- UI checks ran on Chromium only (formal browser list: SAD OI-02, *To be confirmed*).
- Open design items (not defects): no login rate limiting (OI-04); anyone can register as an Advisor (OI-05).

8. Conclusion

All 59 system test cases and 5 supplementary checks passed; all 53 API operations were exercised; every functional, non-functional and security requirement is validated; no open defects. The exit criteria of the Software Test Plan v1.1 (§5) are met.

Appendix – test case results

TC	Requirement	Priority	Objective	Actual result	Result
TC-001	FR-01	High	Register a new Individual account with valid data.	HTTP 201; role individual; no password/hash in body: true; ww_session cookie set: true	**Pass**
TC-002	FR-01	High	Reject registration with an email that already exists.	HTTP 409 "An account with that email already exists."; user rows unchanged: true; cookie set: false	**Pass**
TC-003	FR-01	High	Verify the minimum password length boundary.	(a) HTTP 422 "Password must be at least 8 characters"; (b) HTTP 201	**Pass**
TC-004	FR-01	Medium	Reject an invalid PAN format and an invalid email.	(a) HTTP 422 panNumber: "PAN must look like ABCDE1234F"; (b) HTTP 422 email: "Invalid email"	**Pass**
TC-005	FR-02, SR-02	High	Log in with valid credentials and verify session cookie attributes.	HTTP 200; cookie flags: Path=/, Expires=Mon, 09 Nov 2026 14:17:10 GMT, Max-Age=604800, Secure, HttpOnly, SameSite=lax; JWT alg HS256, iss wealthwise, sub = user id: true, exp present: true	**Pass**
TC-006	FR-02, SR-03	High	Verify login failures do not reveal whether an account exists.	All 10 → 401 "Invalid email or password.", no cookie: true; mean 358 ms (existing) vs 417 ms (unknown)	**Pass**
TC-007	FR-03	High	Log out and confirm the session ends.	me before: priya@wealthwise.dev; logout HTTP 200 {"ok":true}, cookie cleared: true; me after: {"user":null}; /api/assets → 401	**Pass**
TC-008	FR-04, SR-04	High	Verify protected APIs and pages require authentication.	6 APIs → 401 "Authentication required.": true; /dashboard → HTTP 307 to /login	**Pass**

TC	Requirement	Priority	Objective	Actual result	Result
TC-009	FR-04, SR-04	High	Verify advisor-only functions are refused to Individual users.	/api/advisor/clients → 403 "Advisor role required."; /api/news/refresh → 403 "Advisor role required."	**Pass**
TC-010	FR-05, FR-18	High	Create an asset and verify it is listed and reflected in net worth.	HTTP 201; liquidity defaulted to 'liquid'; in list; total assets +105000, net worth +105000	**Pass**
TC-011	FR-05, SR-06	High	Reject a valuation date earlier than the purchase date.	HTTP 422; valuationDate: "Valuation date cannot be before the purchase date"; no asset created: true	**Pass**
TC-012	FR-05	Medium	Verify value boundaries for assets.	(a) 201; (b) 422 currentValue error: true; (c) 422 assetType error: true	**Pass**
TC-013	FR-06	Medium	Revaluing an asset appends to its valuation history.	PUT HTTP 200; history: 2025-04-01 105000 (manual-entry) → 2026-11-02 110000 (revaluation)	**Pass**
TC-014	FR-05	Medium	Deleting an asset keeps linked ledger history.	DELETE 200 {"ok":true}; GET → 404; transaction kept: true, relatedAssetId = null	**Pass**
TC-015	FR-07, NFR-02	High	Create a loan and verify the generated EMI schedule is exact.	HTTP 201; 240 installments; EMI ₹43391.16; Σ principal ₹5000000; final balance 0; liabilities +5000000	**Pass**
TC-016	FR-07, SR-06	High	Verify interest-rate and tenure boundaries.	rate 0 → 422; rate 60.01 → 422; rate 0.01 → 201; rate 60 → 201; tenure 0 → 422; tenure 481 → 422; tenure 1 → 201; tenure 480 → 201; installments for tenure 1/480: 1/480	**Pass**
TC-017	FR-08	Medium	View loan list progress and amortisation schedule.	2 loans each with EMI, paid count, outstanding, next due date: true; HDFC Bank schedule 60 rows with all columns; detail page HTTP 200	**Pass**
TC-018	FR-08	High	Editing the terms of a loan with payments is refused.	HTTP 409 "This loan already has paid installments, so its terms cannot be changed."; schedule unchanged: true	**Pass**
TC-019	FR-09	High	Mark an EMI paid atomically.	HTTP 200 amount 43391.16; status paid on 2026-11-02; loan ledger entries 27 → 28 ('EMI Payment'); outstanding fell by 7974.49 (principal component 7974.49)	**Pass**
TC-020	FR-09	High	Refuse paying the same installment twice.	HTTP 404 "Installment not found, not yours, or already paid."; ledger unchanged: true	**Pass**
TC-021	FR-09	High	Paying the last unpaid installment closes the loan.	pay HTTP 200, loanClosed true; loan status 'closed'; in dues: false	**Pass**
TC-022	FR-10	High	Past-due installments are marked overdue and bucketed.	markedOverdue 1; installment due 14 days ago → overdue/overdue (days until due -14); due in 3 days → due_this_week	**Pass**
TC-023	FR-11	Medium	Simulate a prepayment without changing data.	HTTP 200; reduce tenure saves ₹1422060.03 (213→169 months), reduce EMI saves ₹470090.24 (EMI 43391.16→38836.75); stored schedule unchanged: true	**Pass**
TC-024	FR-11	Medium	Apply a prepayment funded from an asset; net worth unchanged.	HTTP 201; outstanding 4763640.39 → 4463640.39; FD 500000 → 200000; net worth 5287014.37 → 5287014.37 (unchanged: true); prepayments recorded: 1	**Pass**
TC-025	FR-11, NFR-03	High	Reject over-payment and roll back an under-funded prepayment.	(a) 422 "Amount must be less than the outstanding principal."; (b) 422 "The funding asset is worth less than the prepayment."; loan, schedule and asset unchanged: true	**Pass**

TC	Requirement	Priority	Objective	Actual result	Result
TC-026	FR-12	High	Record a transaction and retrieve it with filters.	POST 201; found by filter: true; DELETE 200; gone: true	**Pass**
TC-027	FR-12, SR-06	High	Validate transaction amount and date.	amount 0 → 422; -5 → 422; 0.01 → 201; date '01-10-2026' → 422 "Expected YYYY-MM-DD"; type 'gift' → 422	**Pass**
TC-028	FR-13	Medium	Full-text search matches stems and phrases.	"pharmacy" → rows including "Apollo Pharmacy" (stem match); "\"indian oil\"" → 12 rows, every one containing the phrase "Indian Oil"; the row "Oil and Indian spices" was not returned	**Pass**
TC-029	FR-14, FR-15	High	Preview an HDFC-format statement.	HTTP 200; bank HDFC; 3 rows (salary 165000 income on 2026-10-01; Swiggy → Food); rejected: Invalid date "31/02/26"; Missing description; ledger unchanged: true	**Pass**
TC-030	FR-14	High	Commit an import, then re-commit the same rows (de-duplication).	first: 201 inserted 3, duplicates 0; second: 201 inserted 0, duplicates 3; batches listed: 2	**Pass**
TC-031	FR-14, SR-07	High	Reject invalid statement files.	(a) 415 "Only .csv files can be imported."; (b) 422 "Could not find a header row with Date, Description/Narration and amount columns."; (c) 413 "The file is larger than 5 MB."; (d) 422 "Could not find a header row with Date, Description/Narration and amount columns."	**Pass**
TC-032	FR-14, NFR-03	High	A failure mid-batch leaves no rows inserted.	HTTP 422 "A value is not valid (for example an impossible date or a number out of range)." (2nd row has an impossible date the database rejects); transactions 3 → 3; batch rows written: 0. Also covered by tests/db/imports.test.ts and atomicity.test.ts	**Pass**
TC-033	FR-15	Medium	User rule precedence and retroactive re-categorisation.	POST 201, recategorised 1; user 2 row 'Business Expense'; user 3 row 'Shopping'; delete system rule → 404	**Pass**
TC-034	FR-16	Medium	Scan a receipt image without saving data.	HTTP 200 in 1.0 s; merchant 'APOLLO PHARMACY', amount 455.28 (Grand Total; Sub Total is 435.00), date 2026-10-12, confidence 94%, category 'Health'; ledger unchanged: true	**Pass**
TC-035	FR-16, SR-07	Medium	Confirm receipt; refuse duplicate and invalid uploads.	confirm 201 (filed in vault: true); repeat 409 "This receipt is already recorded."; GIF 415; 9 MB JPEG 413	**Pass**
TC-036	FR-17	Medium	Upload, find and delete a vault document.	POST 201; stored as b7cbe1d9-4bbc-43a3-baac-8d18b3d13847.pdf; found by tag and text: true; stats count 1; DELETE 200; gone: true	**Pass**
TC-037	FR-17, SR-07	High	Reject disallowed vault uploads.	(a) .exe 415; (b) .html 415; (c) 10 MB+1 413; (d) '../../evil.pdf' 201, stored as 3a57917f-8d04-4b7d-8106-4d2235bf9649.pdf inside uploads/: true (original name kept as metadata '../../evil.pdf')	**Pass**
TC-038	FR-17, NFR-04	Medium	Vault reports unavailability when its file storage cannot be written.	tests/db/vault.test.ts "vault reports unavailable (503) when storage cannot be written": passed – POST /api/vault returns 503 "The document vault is unavailable right now…"	**Pass**
TC-039	FR-18	High	Dashboard data is complete and consistent; new users see zeros.	priya: assets 10407000 = Σ assets 10407000; liabilities 5119985.63 = Σ outstanding 5119985.63; allocation Σ 100.0 %; dues 4; cashflow 12 months; recurring 7 (all ≥ 3 months); new user HTTP 200 with zero totals	**Pass**
TC-040	FR-19	Medium	Derived credit score range, factors, disclaimer and reaction to overdue EMI.	score 571 (Fair) with 5 factors (Payment history 35%, Debt to assets 25%, Loan mix 15%, Length of history 15%, Track record 10%); disclaimer "Indicative score derived from your WealthWise records only. It is not a CIBIL, Experian or Equifax credit score."; after one overdue installment: 545	**Pass**

TC	Requirement	Priority	Objective	Actual result	Result
TC-041	FR-20	Low	Loan recommendations rank eligible products and explain ineligibility.	HTTP 200; score 767; 2 eligible ranked #1 Home Loan Classic 9.04%, #2 Home Loan Advantage 9.12%; ineligible: Easy Home (Maximum tenure is 300 months)	**Pass**
TC-042	FR-21	Low	What-if projection results and parameter limits.	(a) 200; scenario gain at month 60 ₹127823.8 vs remaining interest ₹127823.8; (b) months=5 → 422; (c) another user's loan → 404 "Loan not found."	**Pass**
TC-043	FR-22	Medium	Financial-year boundary handling.	FY 2025-26: expense 100 (31 Mar only); FY 2026-27: expense 200 (1 Apr only); sections summary, category spend, loans, tax hints present: true	**Pass**
TC-044	FR-23, SR-09	Medium	Download the PDF statement and verify PAN masking.	HTTP 200; application/pdf; attachment; filename="wealthwise-statement-2026-11-02.pdf"; Cache-Control no-store; sections present: true; PAN shown as XXXXXX234K, full PAN absent: true; amounts as "Rs. 1,07,07,000.00"	**Pass**
TC-045	FR-24	Medium	Reminders are generated once and can be marked read.	after two reads: 2 reminders (due_soon,overdue), unread 2; mark one read → 1; mark all → 0	**Pass**
TC-046	FR-25	Low	News feed, rate impact and fallback.	feed source 'sample' (RBI feed unreachable from the test network), 3 articles, all labelled illustrative: true; +25 bps impact on 1 floating loan(s): SBI EMI 43368.4 → 44029.57; non-rate article: "This article does not describe a lending-rate change."; advisor refresh → 503 "The RBI news feed is unavailable (fetch failed)."	**Pass**
TC-047	FR-26, SR-05	High	Advisor sees only assigned clients.	clients: Priya Sharma, Rahul Verma (each with net worth summary); priya's detail 200 with 5 assets; non-assigned user → 403 "This user is not your client."; rahul calling the advisor API → 403 "Advisor role required."	**Pass**
TC-048	SR-01	High	Passwords are stored only as bcrypt cost-12 hashes and never returned.	stored hash starts '$2a$12$' (60 chars); password or hash found in responses or server log: false	**Pass**
TC-049	SR-02	High	Tampered, wrongly signed or expired tokens are rejected.	(a) tampered, (b) other secret, (c) expired, (d) alg none → /api/assets 401/401/401/401; /api/auth/me user null/null/null/null	**Pass**
TC-050	SR-05	High	Cross-user object access is refused.	loan GET, EMI pay, asset PUT, asset DELETE, transaction DELETE, prepay, vault DELETE, vault file download → 404/404/404/404/404/404/404/404; priya's data unchanged: true	**Pass**
TC-051	SR-06	High	Injection payloads are treated as data.	search payload → 200 with 0 rows; DROP TABLE payload → 200 with 0 rows; error details leaked: false; transactions table intact (+1 row from the script-text entry); script description stored as plain text (React escapes it on the ledger page)	**Pass**
TC-052	SR-08	Medium	Unexpected errors return a generic message.	HTTP 500 body {"error":"Internal server error"}; connection details or stack in body: false; detail written to the server log: true	**Pass**
TC-053	SR-09, NFR-07	Medium	Secrets come only from the environment.	31-character secret: login HTTP 500, no cookie, server log "JWT_SECRET must be set and at least 32 characters long"; hard-coded secrets / connection strings in src or scripts: none; .env ignored by Git: true	**Pass**
TC-054	NFR-01	Medium	Measure dashboard and OCR response times.	dashboard p90 9 ms over 20 requests (max 13 ms); OCR scans 0.8 s, 0.7 s, 0.8 s, 0.8 s, 0.8 s	**Pass**

TC	Requirement	Priority	Objective	Actual result	Result
TC-055	NFR-02	High	Exact arithmetic for edge-case schedules.	(a) EMI for ₹1,20,000 at 0 % over 12 months = ₹10000, no division error; (b) ₹1,00,001 at 7.25 % for 37 months: 37 installments, Σ principal ₹100001, final balance 0	**Pass**
TC-056	NFR-04	High	Core features keep working when an optional dependency fails, without stalling.	with the RBI feed unreachable: asset, loan, EMI payment, transaction, dashboard, import preview and commit → 201/201/200/201/200/200/201; news fell back to cached/sample articles (3 ms, then 3 ms – failed fetch not retried within 60 s); vault storage failure → 503 covered by TC-038	**Pass**
TC-057	NFR-05	Medium	Theme, number formatting and chart accessibility.	dark theme kept after reload: true; dark class set before first paint: true; amounts like ₹52,87,014.37: true; every chart has a legend, labels or table: true; ledger shows "<script>" text without running it: true	**Pass**
TC-058	NFR-06	Medium	Type safety, automated tests and layering.	typecheck: 0 errors; unit tests: 73 passed, 0 failed; database tests: 54 passed, 0 failed; layering checked by tests/unit/architecture.test.ts (part of the unit run)	**Pass**
TC-059	NFR-07	Medium	Fresh installation from documented steps.	Fresh checkout → npm install → npm run db:migrate → npm run db:seed → npm run build → npm start completed without code changes; app served on the chosen port (/login HTTP 200); demo login works (TC-005). Same steps run in GitHub Actions CI on a clean runner for every PR	**Pass**

Sign-off

Role	Name	Signature / Date
Test Lead	Likith Kumar	
Security & Traceability Lead	Vasu Tripathi	
Architecture Lead	Nishant Nag	
Requirements / UAT Lead	Madhusudhan P	